Outlook

## Re: HMMasting

**From** Michael Betancourt <betan@symplectomorphic.com>
**Date** Tue 5/26/2026 1:45 PM
**To** Van der Meersch, Victor <victor.vandermeersch@ubc.ca>

[**CAUTION:** Non-UBC Email]

Victor,

Apologies for the delay.

You're right — I do think that many of the sentences run on for way too long. I notice that there are a lot of parenthetical clauses; when added to a simple sentence they're fine, but they're often added to already complicated sentences which makes it hard to keep all of the clauses organized. Personally I prefer to avoid actual parentheses as much as possible, using commas, and occasionally em-dashes, for parenthetical clauses. Slightly more objectively, I don't think using single em-dashes to separate clauses is well-regarded. I think these would all be better as commas, if not semi-colons or colons when appropriate. That, however, is all style. I'm happy to comment further or offer edits if/when it gets to that point.

Overall the structure looks good. I have mostly small comments here and there.

Line 14:
Maybe replace "a population" with "target population" or something else more precise?

Line 23:
Maybe "continued synchrony for the immediate future" or the like?

Figure 1:
Personally, I would break this down into three separate figures: one for A, one for B, and C (which would also allow you to stack them next to each other horizontally instead of vertically), and one for D.

Line 84:
"Test" is a dangerous word when statistics are involved. I would use a synonym with less connotations, such as "study" or "investigate" just to avoid the risk of triggering a reviewer.

Line 101:
It's not just some zero seed counts, it's _a lot_ of them, no? Maybe "characterized by extremely low counts, including a high probability of no observed seeds at all."

Lines 110-112:
This reads pretty awkwardly, as it sounds like you're saying that trees are both likely to transition but also likely to not transition. Maybe something like "Trees commonly shift into the opposite reproductive state each year, but the low reproductive state is much more persistent. On the

other hand, trees rarely stay in the high reproductive state for more than a single year".

Figure 2C:
"Random" can be another trigger word.  Perhaps "arbitrary tree"?

Lines 117-120:
I think this is worth a plot comparing the two probability posteriors!

Figure 3:
The final sentence of the caption seems deprecated: there's no red lines, and no gray box plots anywhere.

Lines 172-175:
I would love to see a figure combining Figure 3A along with a corresponding figure for the high -> high transition probability summer temperature dependence.

Figure 5:
Caption is missing a sentence or two explaining the relevance of this plot.  Is the point that the between population synchrony is slightly, but systematically, smaller than the within population synchrony?

Figure 6:
What's the x-axis?  Is this a hypothetical?

Lines 257-259:
What are the baselines?

-Michael

> On May 10, 2026, at 6:41 AM, Van der Meersch, Victor <victor.vandermeersch@ubc.ca> wrote:
>
> Hi Mike,
>
> We've been working with Lizzie on a manuscript related to HMM and masting.
> I'm still finalizing some stuff (abstract, references, appendix), but I believe it would be nice to have your feedback at this point.
> (Disclaimer: some sentences might be too long and need to be broken up)
>
> Please find attached the pdf. I can send you the .tex or .Rnw file if you prefer.
> I think the goal is to submit at the end of the month, but no worries if you need more time.
> Thank you for your help on this!
>
> All the best,
> Victor
> <beechmasting_ms.pdf>